LAB 17.2 Sequence-to-Sequence Tasks — Using Hugging Face Pipelines for Translation

STEP 1 — Install and Import Required Libraries

```
!pip install transformers
!pip install torch
from transformers import pipeline
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.6.0
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
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Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hugg
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Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

Explanation

Pretrained libraries are used in machine translation because they already contain large amounts of trained knowledge from massive datasets. This saves time and computational cost. Advantages of Transformers Better context understanding Parallel processing Higher accuracy Handles long sentences better than traditional models like RNN or LSTM.

STEP 2 — Load Translation Model

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# English to French
tokenizer_fr = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-fr")
model_fr = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en-fr")

def translate_en_to_fr(text):
    inputs = tokenizer_fr(text, return_tensors="pt")
    outputs = model_fr.generate(**inputs)
    return tokenizer_fr.decode(outputs[0], skip_special_tokens=True)

# English to German
tokenizer_de = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-de")
model_de = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en-de")

def translate_en_to_de(text):
    inputs = tokenizer_de(text, return_tensors="pt")
    outputs = model_de.generate(**inputs)
```

```
    return tokenizer_de.decode(outputs[0], skip_special_tokens=True)

# English to Hindi
tokenizer_hi = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-hi")
model_hi = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en-hi")

def translate_en_to_hi(text):
    inputs = tokenizer_hi(text, return_tensors="pt")
    outputs = model_hi.generate(**inputs)
    return tokenizer_hi.decode(outputs[0], skip_special_tokens=True)

# Example
text = "Hello, how are you?"

print(translate_en_to_fr(text))
print(translate_en_to_de(text))
print(translate_en_to_hi(text))
```

```
tokenizer_config.json: 100%  42.0/42.0 [00:00<00:00, 698B/s]

source.spm: 100%  778k/778k [00:00<00:00, 8.53MB/s]

target.spm: 100%  802k/802k [00:00<00:00, 9.84MB/s]

vocab.json:  1.34M/? [00:00<00:00, 14.7MB/s]

/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip ins
  warnings.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: 100%  301M/301M [00:03<00:00, 147MB/s]

model.safetensors: 100%  301M/301M [00:04<00:00, 138MB/s]

Loading weights: 100%  258/258 [00:00<00:00, 429.02it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu

generation_config.json: 100%  293/293 [00:00<00:00, 2.36kB/s]

config.json:  1.33k/? [00:00<00:00, 11.2kB/s]

tokenizer_config.json: 100%  42.0/42.0 [00:00<00:00, 1.31kB/s]

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target.spm: 100%  797k/797k [00:00<00:00, 6.36MB/s]

vocab.json:  1.27M/? [00:00<00:00, 32.2MB/s]

pytorch_model.bin: 100%  298M/298M [00:02<00:00, 155MB/s]

model.safetensors: 100%  298M/298M [00:04<00:00, 110MB/s]

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config.json:  1.39k/? [00:00<00:00, 19.8kB/s]

tokenizer_config.json: 100%  44.0/44.0 [00:00<00:00, 3.75kB/s]

source.spm: 100%  812k/812k [00:00<00:00, 38.3MB/s]

target.spm: 100%  1.07M/1.07M [00:00<00:00, 47.4MB/s]

vocab.json:  2.10M/? [00:00<00:00, 43.8MB/s]

pytorch_model.bin: 100%  306M/306M [00:06<00:00, 43.0MB/s]

model.safetensors: 100%  306M/306M [00:06<00:00, 67.1MB/s]

Loading weights: 100%  258/258 [00:03<00:00, 100.26it/s, Materializing param=model.shared.weight]

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The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu

generation_config.json: 100%  293/293 [00:00<00:00, 18.4kB/s]

Bonjour, comment allez-vous ?
Hallo, wie geht's?
हैलो, तुम कैसे हो?
```

Explanation

A translation pipeline is a ready-to-use interface provided by Hugging Face that performs tasks such as text translation using pretrained models. Pretrained models are suitable because they are already trained on large multilingual datasets.

STEP 3 — Prepare Input Sentences

```
sentences = [
"Artificial Intelligence is transforming the world.",
"I love learning new technologies.",
"The weather is very pleasant today.",
"Machine learning helps computers learn from data.",
"Can you help me with this problem?",
"She completed her project before the deadline.",
"The conference was attended by many researchers.",
"Technology is growing rapidly in modern society.",
"He bought fruits and vegetables from the market.",
"Education plays an important role in development.",
"If you work hard, you will succeed.",
"The student asked a question during the lecture.",
"This research focuses on natural language processing.",
"Social media connects people across the world.",
"Do you know how machine translation works?"
]
```

Explanation

Using different sentence structures helps evaluate the robustness of translation systems. Complex and technical sentences are usually harder to translate.

STEP 4 — Perform Translation

```
for sentence in sentences:
    result = translate_en_to_fr(sentence)
    print("English:", sentence)
    print("French:", result)
    print()
```

```
English: Artificial Intelligence is transforming the world.
French: L'intelligence artificielle transforme le monde.

English: I love learning new technologies.
French: J'adore apprendre les nouvelles technologies.

English: The weather is very pleasant today.
French: Le temps est très agréable aujourd'hui.

English: Machine learning helps computers learn from data.
French: L'apprentissage automatique aide les ordinateurs à apprendre des données.

English: Can you help me with this problem?
French: Pouvez-vous m'aider avec ce problème ?

English: She completed her project before the deadline.
French: Elle a terminé son projet avant la date limite.

English: The conference was attended by many researchers.
French: De nombreux chercheurs ont assisté à la conférence.

English: Technology is growing rapidly in modern society.
French: La technologie se développe rapidement dans la société moderne.

English: He bought fruits and vegetables from the market.
French: Il a acheté des fruits et légumes du marché.

English: Education plays an important role in development.
French: L'éducation joue un rôle important dans le développement.

English: If you work hard, you will succeed.
French: Si vous travaillez dur, vous réussirez.

English: The student asked a question during the lecture.
French: L'étudiant a posé une question au cours de la conférence.

English: This research focuses on natural language processing.
French: Cette recherche se concentre sur le traitement du langage naturel.

English: Social media connects people across the world.
French: Les médias sociaux relient les gens à travers le monde.

English: Do you know how machine translation works?
French: Savez-vous comment fonctionne la traduction automatique?
```

Explanation

The translation model converts the source language (English) into the target language (French) using an encoder-decoder architecture.
Observed pattern: Short sentences translate more accurately compared to long technical sentences.

STEP 5 — Translate into Multiple Languages

```
for sentence in sentences:
    fr = translate_en_to_fr(sentence)
    de = translate_en_to_de(sentence)
    hi = translate_en_to_hi(sentence)

    print("English:", sentence)
    print("French:", fr)
    print("German:", de)
    print("Hindi:", hi)
    print("-----------------------------------------------")
```

```
English: Artificial Intelligence is transforming the world.
French: L'intelligence artificielle transforme le monde.
German: Künstliche Intelligenz verwandelt die Welt.
Hindi: कलावादी तत्व दुनिया को बदल रहा है।
-----------------------------------------------
English: I love learning new technologies.
French: J'adore apprendre les nouvelles technologies.
German: Ich liebe es, neue Technologien zu lernen.
Hindi: मैं नई तकनीक सीखने को प्यार करता हूँ.
-----------------------------------------------
English: The weather is very pleasant today.
French: Le temps est très agréable aujourd'hui.
German: Das Wetter ist heute sehr angenehm.
Hindi: मौसम आज बहुत सुखद है.
-----------------------------------------------
English: Machine learning helps computers learn from data.
French: L'apprentissage automatique aide les ordinateurs à apprendre des données.
German: Maschinelles Lernen hilft Computern, aus Daten zu lernen.
Hindi: मशीन सीखने से कंप्यूटर को डाटा से सीखने में मदद मिलती है.
-----------------------------------------------
English: Can you help me with this problem?
French: Pouvez-vous m'aider avec ce problème ?
German: Können Sie mir bei diesem Problem helfen?
Hindi: आप इस समस्या के साथ मेरी मदद कर सकते हैं?
-----------------------------------------------
English: She completed her project before the deadline.
French: Elle a terminé son projet avant la date limite.
German: Sie hat ihr Projekt vor Ablauf der Frist abgeschlossen.
Hindi: उसने अपनी परियोजना को मृतलाइन से पहले पूरा किया.
-----------------------------------------------
English: The conference was attended by many researchers.
French: De nombreux chercheurs ont assisté à la conférence.
German: An der Konferenz nahmen viele Forscher teil.
Hindi: कई खोजकर्ताओं ने सम्मेलन में हाज़िर होने का फैसला किया ।
-----------------------------------------------
English: Technology is growing rapidly in modern society.
French: La technologie se développe rapidement dans la société moderne.
German: Die Technologie wächst in der modernen Gesellschaft rasant.
Hindi: आधुनिक समाज में प्रौद्योगिकी तेज़ी से बढ़ती जा रही है ।
-----------------------------------------------
English: He bought fruits and vegetables from the market.
French: Il a acheté des fruits et légumes du marché.
German: Er kaufte Obst und Gemüse vom Markt.
Hindi: वह बाजार से फल और सब्ज़ियाँ खरीदा.
-----------------------------------------------
English: Education plays an important role in development.
French: L'éducation joue un rôle important dans le développement.
German: Bildung spielt eine wichtige Rolle in der Entwicklung.
Hindi: विकास में शिक्षा एक महत्त्वपूर्ण भूमिका अदा करती है ।
-----------------------------------------------
English: If you work hard, you will succeed.
French: Si vous travaillez dur, vous réussirez.
German: Wenn du hart arbeitest, wirst du Erfolg haben.
Hindi: अगर आप मेहनत करते हैं, तो आप ज़रूर कामयाब होंगे ।
-----------------------------------------------
English: The student asked a question during the lecture.
French: L'étudiant a posé une question au cours de la conférence.
German: Der Student stellte während des Vortrags eine Frage.
```

Explanation

Languages have different grammar rules and sentence structures. Example differences: English follows Subject-Verb-Object Hindi often follows Subject-Object-Verb German has complex grammatical cases. These differences create challenges for machine translation.

STEP 6 — Translation Quality Analysis

(Example)

The translation model generally preserves the core meaning of sentences across languages. Simple sentences are translated with high accuracy and grammatical correctness. However, complex sentences sometimes show slight structural differences in translated outputs. Technical terms such as machine learning and natural language processing are mostly preserved correctly. In some cases, the translated sentence structure changes due to differences in grammar rules between languages. Hindi translations often place the verb at the end of the sentence. German translations may include compound words that combine multiple meanings. Overall, pretrained translation models produce fluent and understandable results, but they may struggle with idioms, cultural context, or very domain-specific terminology.

STEP 7 — Testing with Complex Sentences

```
complex_sentences = [
"Although the experiment was difficult, the researchers successfully completed it.",
"If artificial intelligence continues to grow, it will transform many industries.",
"How does machine translation help people communicate across languages?",
"The professor explained the concept in detail during the lecture."
]

for sentence in complex_sentences:
    print("English:", sentence)
    print("French:", translate_en_to_fr(sentence))
    print()
```

```
English: Although the experiment was difficult, the researchers successfully completed it.
French: Bien que l'expérience ait été difficile, les chercheurs l'ont menée à bien.

English: If artificial intelligence continues to grow, it will transform many industries.
French: Si l'intelligence artificielle continue de croître, elle transformera de nombreuses industries.

English: How does machine translation help people communicate across languages?
French: Comment la traduction automatique aide-t-elle les gens à communiquer entre les langues?

English: The professor explained the concept in detail during the lecture.
French: Le professeur a expliqué le concept en détail au cours de la conférence.
```

Observations

Meaning is mostly preserved. Sentence structure changes in some languages. Long sentences sometimes become simplified.

Answers to Questions

1. What is a Sequence-to-Sequence model?

A Sequence-to-Sequence (Seq2Seq) model is a neural network architecture that converts one sequence of data into another sequence, such as translating a sentence from one language to another.

2. What is machine translation?

Machine translation is the automatic conversion of text from one language to another using computer algorithms and artificial intelligence.

3. What is an encoder–decoder architecture?

Encoder-decoder architecture consists of two components: Encoder: Converts input text into numerical representations. Decoder: Generates the translated output text from those representations.

4. What is a Hugging Face pipeline?

A Hugging Face pipeline is a high-level API that allows users to easily perform NLP tasks like translation, sentiment analysis, and text generation using pretrained models.

5. Why are pretrained models used in NLP?

Pretrained models save training time, require fewer resources, and provide high accuracy because they are trained on very large datasets.

6. Limitations of automatic translation systems

Difficulty with idioms and slang Cultural context misunderstandings Errors in long or complex sentences Domain-specific terminology issues

7. Real-life applications of machine translation

Google Translate for multilingual communication Automatic subtitle translation for videos